Muscles of the upper limb

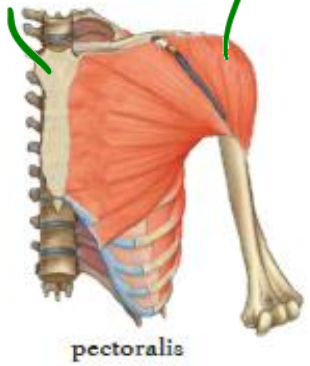


Posterior

sternum

deltoid

shoulder abduction

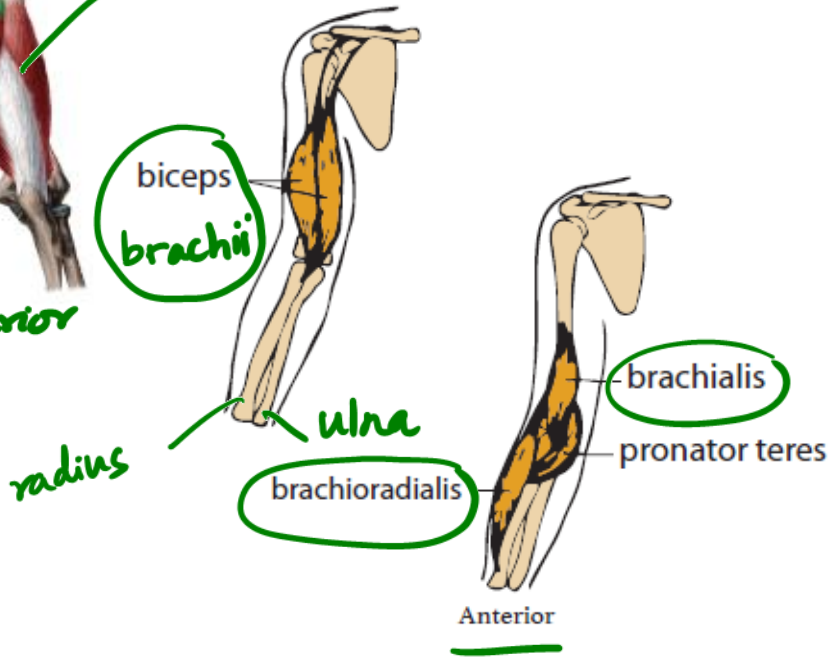


Anterior



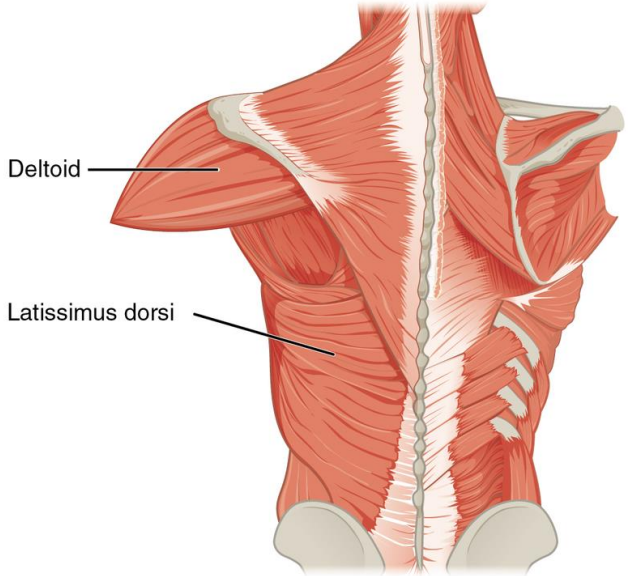
elbow extensors

Posterior



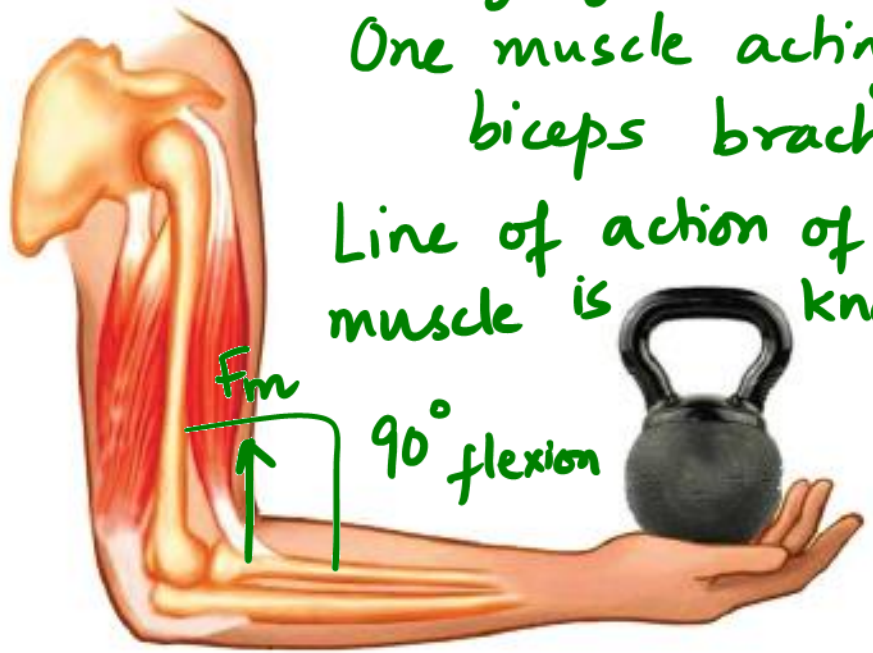
elbow flexors

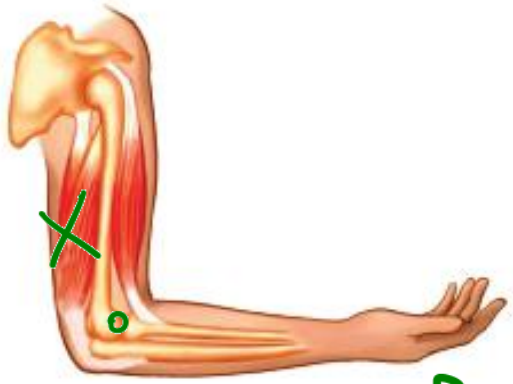
By OpenStax College - https://cnx.org/contents/FPtK1zmh@8.108:y9_gDy74@5, CC BY 3.0, https://commons.wikimedia.org/w/index.php?curid=64291142



Static analysis - elbow

Assumptions: Planar analysis
Hinge joint
One muscle acting - biceps brachii
Line of action of the muscle is known





Unknowns:

F_m, J_x, J_y

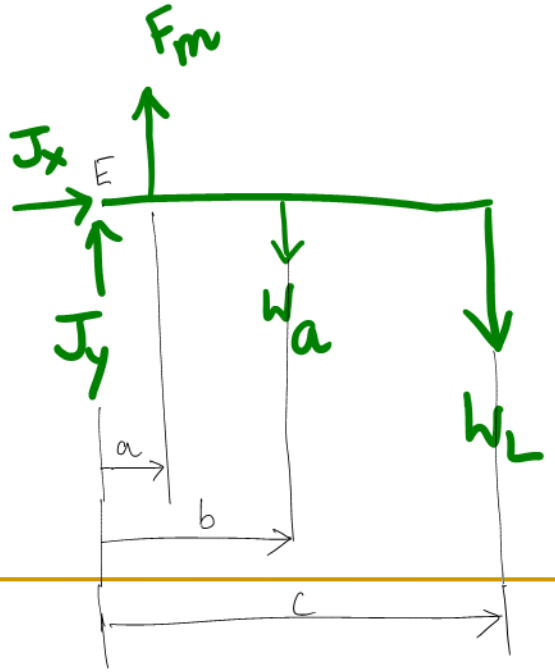
$\Sigma \bar{F} = 0$ / $\Sigma F_x = 0$ / $\Sigma F_y = 0$

$\Sigma M_E = 0$

$+\circlearrowleft \quad F_m a - W_a b - W_L c = 0$

$J_x = 0$

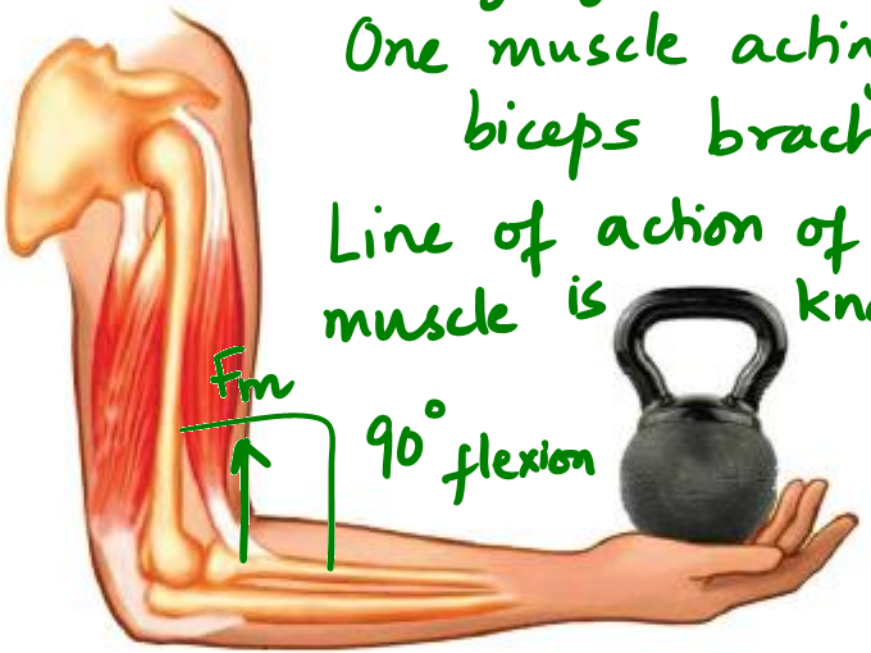
$F_m = \dfrac{W_a b + W_L c}{a}$



$J_y = W_a + W_L - F_m$

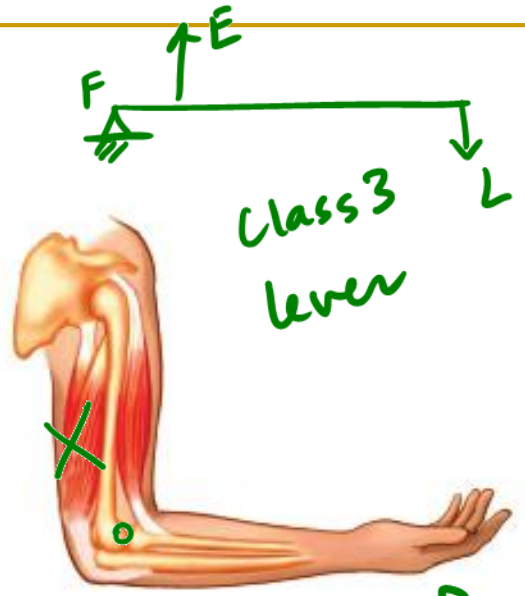
Static analysis - elbow

Assumptions: Planar analysis
Hinge joint
One muscle acting - biceps brachii
Line of action of the muscle is known



$W_a = 20$ N
$W_L = 80$ N

$a = 4$ cm
$b = 15$ cm
$c = 35$ cm

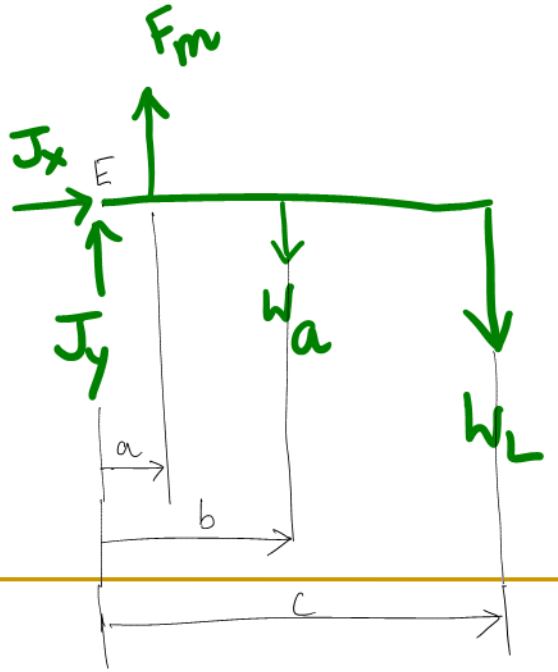


Unknowns:
F_m, J_x, J_y

$\Sigma \bar{F} = 0 \;/\; \Sigma F_x = 0 \;/\; \Sigma F_y = 0$

$\Sigma M_E = 0$

$+\circlearrowleft \quad F_m a - W_a b - W_L c = 0$



$J_x = 0$

$F_m = \dfrac{W_a b + W_L c}{a}$

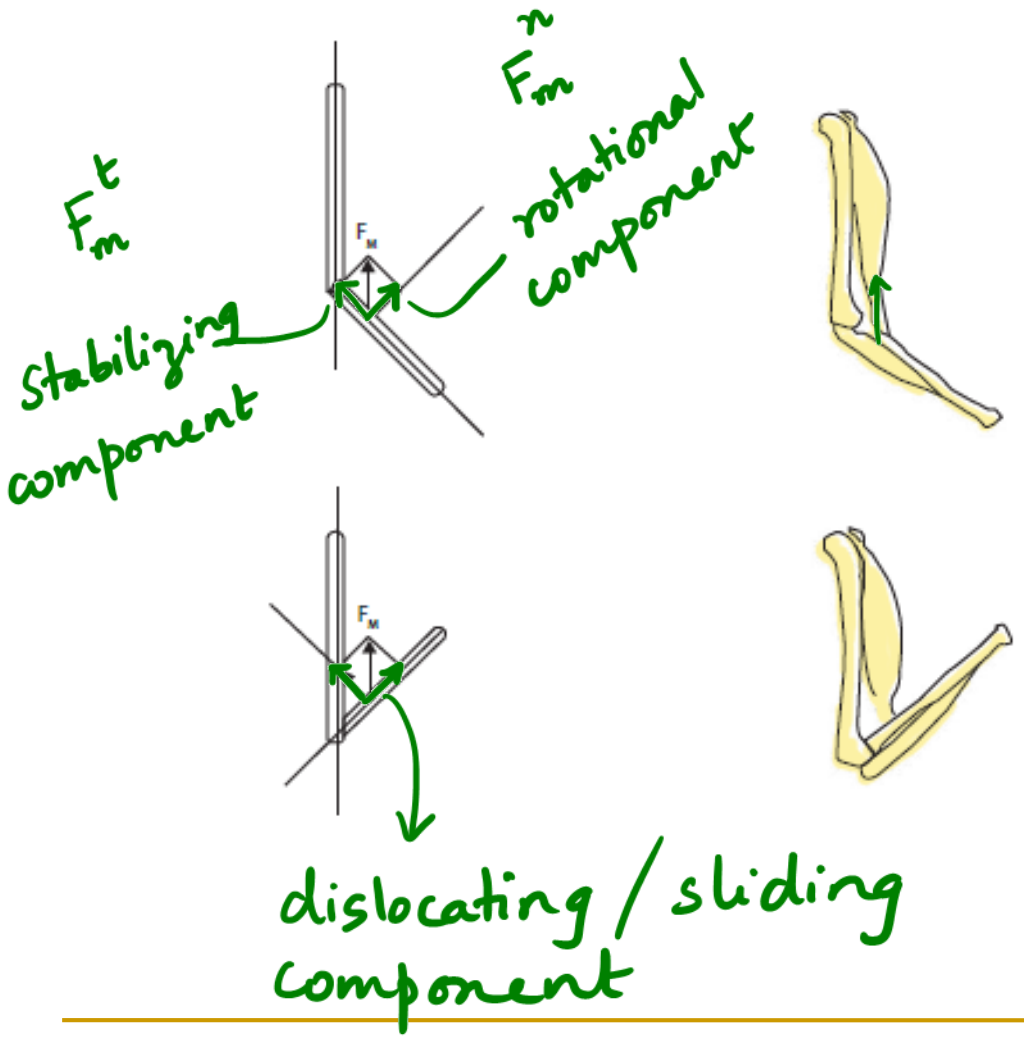
$J_y = W_a + W_L - F_m$

$F_m = 775$ N

$J_x = 0$

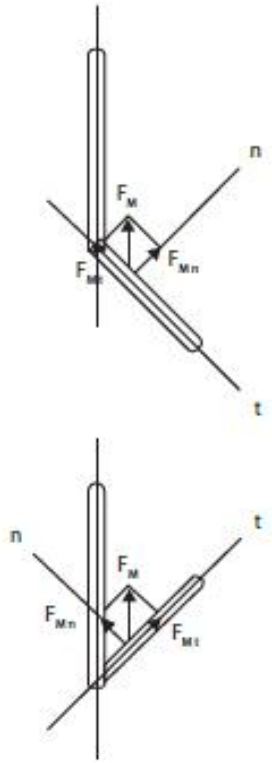
$J_y = -675$ N

Static analysis - elbow



Role of tangential component F_m^t may be stabilizing / dislocating

Static analysis - elbow





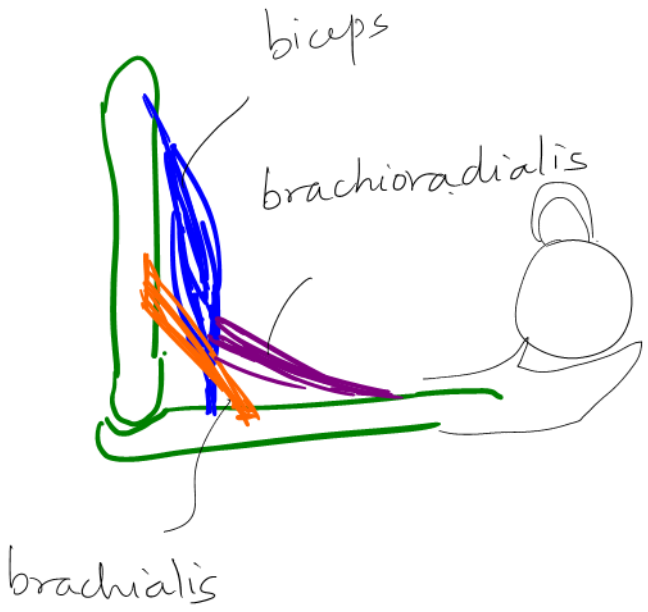




J_y is the resultant force

Elbow is a synovial joint

Joint force result of distributed force applied through the synovial fluid



Assume angles of pull for muscles 1, 2, 3 are known, locations of muscle attachments (insertions) are known, lengths of their moment arms are known

$$\sum F_x = 0$$

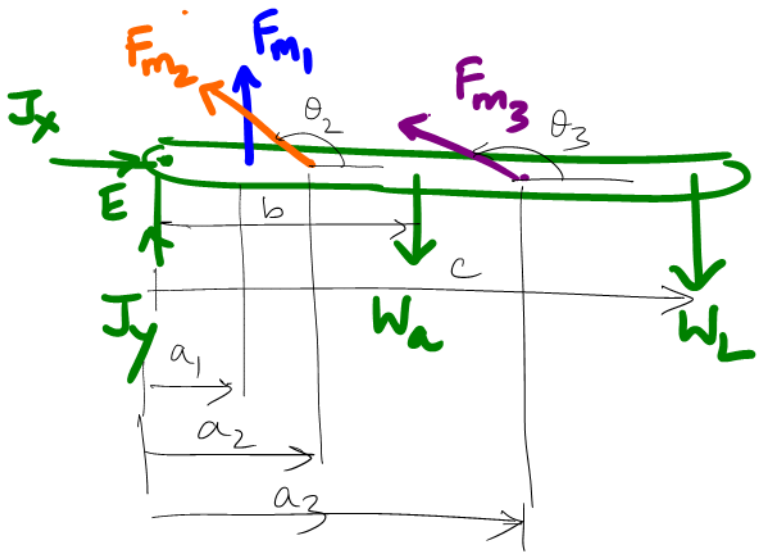
$$J_x + F_{m_2}\cos\theta_2 + F_{m_3}\cos\theta_3 = 0$$

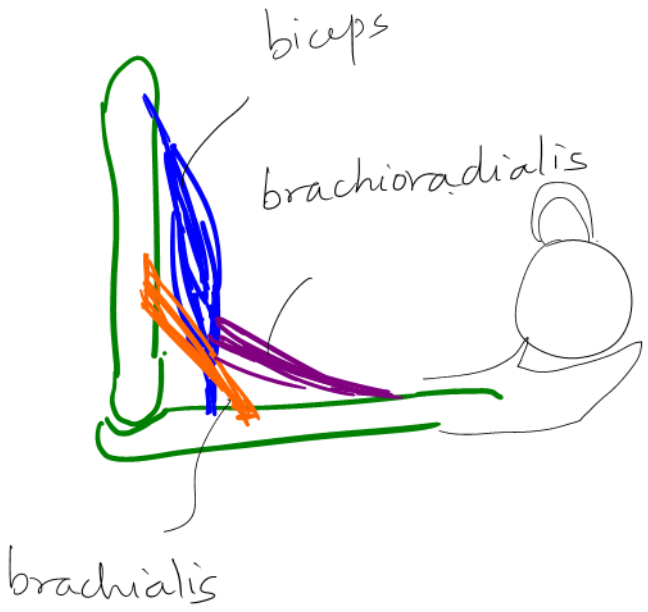
$$\sum F_y = 0$$

$$J_y + F_{m_1} + F_{m_2}\sin\theta_2 + F_{m_3}\sin\theta_3 - W_a - W_L = 0$$

$$\sum M_E = 0$$

$$F_{m_1} a_1 + F_{m_2}\sin\theta_2 \cdot a_2 + F_{m_3}\sin\theta_3 \cdot a_3 - W_a b - W_L c = 0$$





Assume angles of pull for muscles 1, 2, 3 are known, locations of muscle attachments (insertions) are known, lengths of their moment arms are known

$$\sum F_x = 0$$

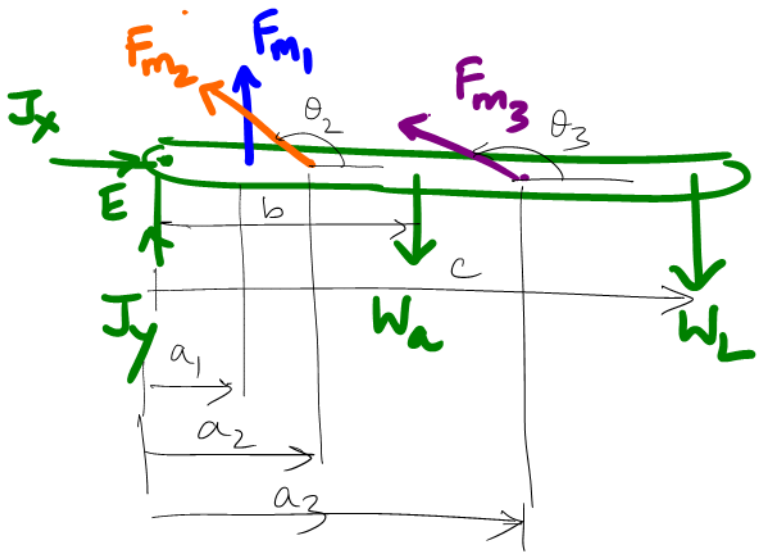
$$J_x + F_{m_2}\cos\theta_2 + F_{m_3}\cos\theta_3 = 0$$

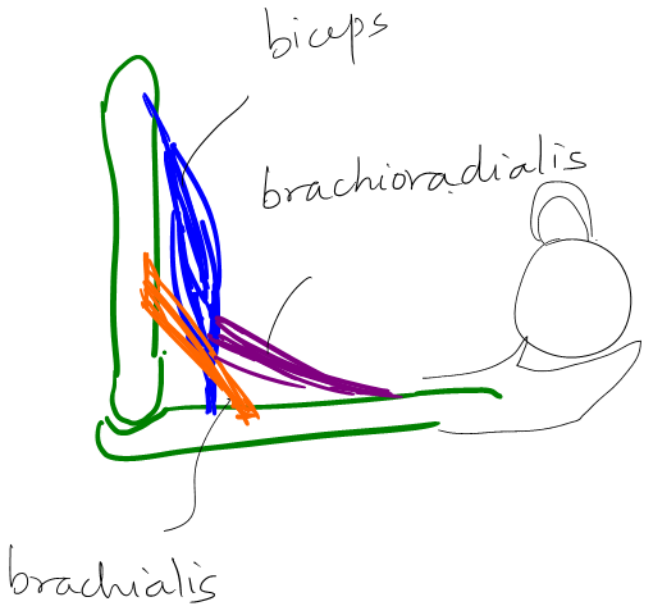
$$\sum F_y = 0$$

$$J_y + F_{m_1} + F_{m_2}\sin\theta_2 + F_{m_3}\sin\theta_3 - W_a - W_L = 0$$

$$\sum M_E = 0$$

$$F_{m_1} a_1 + F_{m_2}\sin\theta_2 \cdot a_2 + F_{m_3}\sin\theta_3 \cdot a_3 - W_a b - W_L c = 0$$



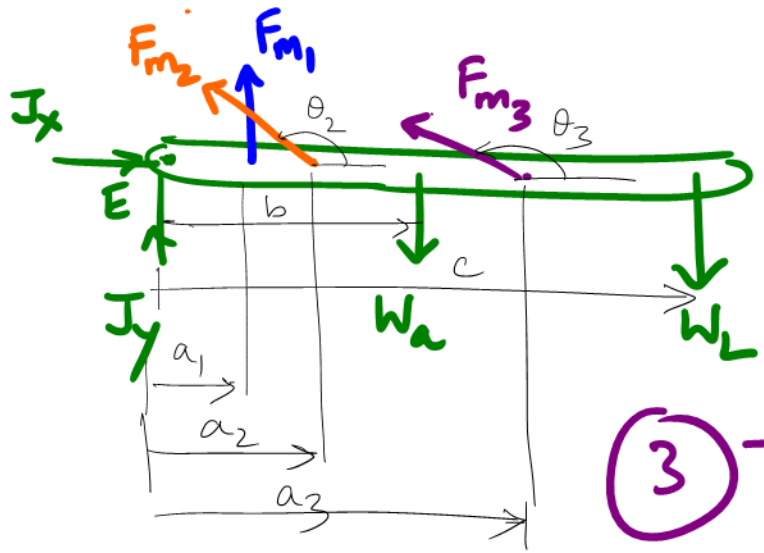


Unknowns:

$F_{m_1}, F_{m_2}, F_{m_3}, J_x, J_y$

Statically indeterminate system

$$\frac{F_{m_2}}{F_{m_1}} = \frac{A_2}{A_1} = k_{21} \qquad \frac{F_{m_3}}{F_{m_1}} = \frac{A_3}{A_1} = k_{31}$$



$$\Sigma F_x = 0$$

(1) — $J_x + F_{m_2}\cos\theta_2 + F_{m_3}\cos\theta_3 = 0$

$$\Sigma F_y = 0$$

(2) — $-J_y + F_{m_1} + F_{m_2}\sin\theta_2 + F_{m_3}\sin\theta_3 - W_a - W_L = 0$

$$\Sigma M_E = 0$$

(3) — $F_{m_1} a_1 + F_{m_2}\sin\theta_2 \cdot a_2 + F_{m_3}\sin\theta_3 \cdot a_3 - W_a b - W_L c = 0$